

RADIAN - RADar-based Intelligent
Activity moNitoring
Matthew Blackwell
Mitch Hoffman
Wyatt Johnson

CONCEPT OF OPERATIONS

CONCEPT OF OPERATIONS FOR RADIAN

TEAM 60

APPROVED BY:

Matthew Blackwell Date

Prof. Jang Date

Roman Venegas Date

Change Record

Rev	Date	Originator	Approvals	Description
1	09/09/25	Matthew Blackwell		Baseline Release
2	09/20/25	Matthew Blackwell		Microcontroller Switch
3	10/04/25	Matthew Blackwell		MCU removal, SMS removal, and other edits from ICD
4	12/05/25	Wyatt Johnson		Document formating

Table of Contents

Table of Contents	3
List of Figures	4
1. Executive Summary	5
2. Introduction	6
2.1. Background	6
2.2. Overview	6
2.3. Referenced Documents and Standards	6
3. Operating Concept	8
3.1. Scope	8
3.2. Operational Description and Constraints	8
3.3. System Description	8
3.3.1. Physical Design	8
3.3.2. Wireless Design	9
3.3.3. Data Design	9
3.4. Modes of Operations	9
3.5. Users	9
3.6. Support	10
4. Scenario(s)	11
4.1. Fall Detection	11
4.2. Normal Activity Monitoring	11
4.3. Multi-Room Monitoring	11
5. Analysis	12
5.1. Summary of Proposed Improvements	12
5.2. Disadvantages and Limitations	12
5.3. Alternatives	12
5.4. Impact	12

List of Figures

Figure 1: High-Level Block Diagram for RADIAN system	6
--	---

1. Executive Summary

Falls are a leading cause of injury among elderly individuals, often resulting in severe health consequences. Existing monitoring solutions on the market, such as wearable devices and camera-based systems, suffer from significant drawbacks. Wearable devices rely on user compliance, and camera-based systems compromise privacy. To address these drawbacks, this project proposes the development of a low-cost, AI-enabled fall detection system using a millimeter wave (mmWave) radar marketed for the elderly at risk of falls.

The RADIAN system uses a commercially available mmWave radar sensor to continuously monitor human activity within an indoor environment, such as a living room or bedroom. The use of mmWave technology allows for privacy to be preserved while still providing adequate protection. The data from the radar will be processed through an embedded microcontroller on the mmWave Radar board and then passed to a Raspberry Pi, where AI algorithms will classify motion patterns to detect falls. In the event of a fall, the RADIAN system will automatically send a wireless alert to a caregiver's smartphone via Wi-Fi, as well as set off an alarm on the device itself to alert anyone inside the house. The budget approved for the RADIAN group is \$375, keeping the project low-cost.

2. Introduction

The RADIAN (RADar-based Intelligent Activity moNitoring) system will detect human falls within an enclosed space, specifically in a household or office setting, using radar-based systems. The RADIAN system will detect a fall using AI-based machine learning algorithms and send an alert to a cellular device, ensuring that the fall is caught quickly and accurately.

2.1. Background

The leading cause of injury-related death among adults ages 65 and older is falls [1]. Many current solutions rely on vision-based systems or a wearable device, both of which suffer from limitations such as user compliance, privacy concerns, and coverage gaps. A millimeter-wave radar preserves privacy, a non-contact, and reliable means of monitoring human activity indoors.

2.2. Overview

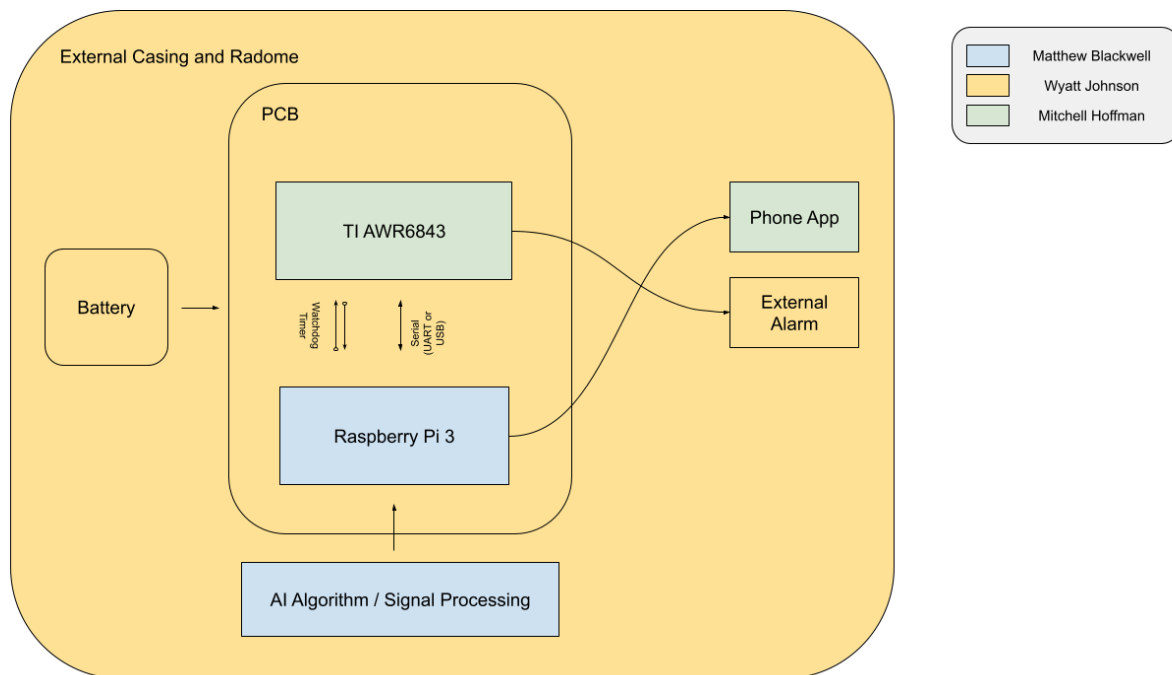


Figure 1: High-Level Block Diagram for RADIAN system

A mmWave Radar Sensor will be used to monitor human movement in a room. Data from the radar is processed using AI algorithms to detect falls in real time. Once a fall is detected, the device will activate an alarm as well as send a remote wireless notification to a mobile device via Wi-Fi.

2.3. Referenced Documents and Standards

- IEEE 802.11 Wi-Fi standards

- IEEE 802.15.4 for wireless communications
 - IEEE 521-2002 Standard Letter Designations for Radar-Frequency Bands
- [1] "Falls Data & Research." Centers for Disease Control and Prevention, Accessed at <https://www.cdc.gov/falls/data-research/index.html#:~:text=Deaths%20from%20older%20adult%20falls,.cdc.gov/STEAD>

3. Operating Concept

3.1. Scope

The RADar-based Intelligent Activity moNitoring (RADIAN) system is designed to enhance the safety of elderly and physically disabled individuals by providing timely alerts in the event of a fall within a monitored room. Once installed, the RADIAN system will continually monitor a closed room for human presence and movement. Fall events are identified by an in-house-developed artificial intelligence (AI) and machine learning algorithm, based on extensive testing by the RADIAN group. When the AI determines that a fall has occurred, a wireless signal will transmit to a dedicated mobile application developed by the RADIAN group, alerting those most important to the monitored individual. In addition, there will be an audible alarm system to alert those in the building to the fall. By combining real-time monitoring with reliable alerting mechanisms, the RADIAN system allows for peace of mind for at-risk individuals and their families.

3.2. Operational Description and Constraints

The RADIAN system is designed to be used in closed rooms of varying sizes, both in room area and ceiling height. The RADIAN system is designed to be relatively easy to use for the user, both in the setup of the radar as well as the alert system and its GUI. A RADIAN system will be set up in the corner of a room, with the sensor and radome facing down at a 45° angle to ensure that the RADIAN system is scanning the entire room. The GUI (Graphical User Interface) will ask what room the RADIAN system is set up in to organize each RADIAN system and to understand what furniture and other objects could be scanned. As the RADIAN system scans the room, the raw data will be processed through the microcontroller embedded in the RADIAN system and then sent to an AI algorithm to analyze the room. If the AI detects a fall, then a signal will be sent to the microcontroller to then alert the GUI that a fall was detected through an app notification.

Here are the resulting constraints of the RADIAN system for it to operate properly according to the description:

- The RADIAN system must not be blocked by any object
- The RADIAN system has to be placed in a high vantage point at a 45° downward angle
- The RADIAN system's radome must be directed towards the largest area of the room
- The RADIAN system must be powered via battery
- The RADIAN system must be small in size (e.g., wall-mountable, no larger than a shoebox) to ensure user-friendliness during installation, thus limiting the complexity of parts that can be used
- The RADIAN group has a budget of \$375, further constraining the complexity of the RADIAN system

3.3. System Description

3.3.1. Physical Design

The radar board, which includes a TI AWR6842ISK and a Raspberry Pi, will all be connected to the battery system on a custom-designed PCB (Printed Circuit Board)

designed in-house by the RADIAN group. This PCB design will enhance the RADIAN system's reliability during installation, as no components can be removed. The RADIAN group will also study and implement the power management in this subsystem, ensuring that the correct voltage and current are being provided to the various components of the RADIAN system. The battery casing will also be screwed to the board and include a removable housing and leads to facilitate battery replacement when needed, ensuring the continued reliability of the RADIAN system.

3.3.2. Wireless Design

The radar chip will be hand-selected through thorough research and trade studies by the RADIAN group to guarantee reliability and affordability for the consumer and users of the RADIAN system. The TI radar board will be programmed to collect the raw data from the radar chip and convert it into usable data for the AI algorithm stored on the Raspberry Pi. Once the AI algorithm has detected a fall, a signal will be sent to an external alarm connected to the RADIAN system. The wireless design subsystem will also be responsible for the application's GUI, providing both a frontend and a backend to ensure that the application is easy to use and runs smoothly.

3.3.3. Data Design

The AI algorithm is the primary focus of the data subsystem, utilizing a simple AI model developed by the RADIAN group to predict fall occurrences. The RADIAN group will do extensive testing and training of the AI model to ensure that false positives and true negatives do not send alerts. The application may have various options to select where the RADIAN system is placed, to help the AI algorithm understand what objects may or may not be in a room. Some examples include a bed present in a bedroom, but not in a living or dining room. Having these distinctions allows for fewer false alarms when the user is lying down vs. falling. The data subsystem will also be responsible for communicating with the cloud server to send the push notifications. A wireless Wi-Fi communication will be sent from the Raspberry Pi 3 to the Gotify cloud to alert those not in proximity of the audible alarm using the RADIAN phone application.

3.4. Modes of Operations

The RADIAN system will only have one operating mode, as the radar has to constantly sweep a room due to the random nature of human movements. Within this mode, the user will be able to define what room the RADIAN system is placed in to aid the AI algorithm in understanding what may or may not be in the room. When operating in the mode, the radar will constantly detect the room and the objects around the room, sensing for motion and a potential fall. If a fall is detected by the RADIAN system and its AI algorithm, then an alert will be sent to the desired device, as well as set off an audible alarm for those in proximity.

3.5. Users

The RADIAN system will be marketed to the high-risk individuals who greatly benefit from a fall detection system in their home without having to worry about privacy infringement with camera-based detection systems. Our product will allow users and their loved ones to have peace of mind when it comes to the health and safety of the user.

The RADIAN system can also double as a simple motion detection device that can be used in various security applications, including home security, business security, and preventing deaths in the back of motor vehicles.

3.6. *Support*

The RADIAN group will provide an extensive manual detailing the installation of the RADIAN system, battery replacement, and a simple explanation of how the RADIAN system works at the highest level. Within the RADIAN application, detailed step-by-step guides will be provided to guide the user through the setup and operation of the device.

4. Scenario(s)

4.1. *Fall Detection*

An elderly person falls in their bedroom or living room. The RADIAN system will use mmWave radar technology to record and process the elderly person's fall. This mmWave radar data will then be sent to an AI model that will then classify the event as a fall, triggering an alarm and notifying the caretaker. This notification will be sent to a caregiver's smartphone through an app. The caregiver will then be able to check on the individual quickly and allow for the necessary response as soon as possible.

4.2. *Normal Activity Monitoring*

The RADIAN system user can be walking, sitting, or lying down in their bedroom. The radar will detect all of these events, but the AI model will classify these events as “safe”. No alerts will be sent in these events, and will allow the user to continuously monitor their environment for hazardous falls without invading their privacy.

4.3. *Multi-Room Monitoring*

The user is capable of monitoring a whole house or a nursing home. The user will install multiple RADIAN systems in each room that they would like to monitor. Each RADIAN system will individually monitor the room in which it is. If the RADIAN system detects a fall, it will notify the caregiver of the event and also provide the location of the room in which it occurred. While one room is triggered, the other rooms will be continuously monitored. All of the data from the rooms can be examined in the RADIAN smartphone app by the caregiver, providing complete protection of the whole building.

5. Analysis

5.1. Summary of Proposed Improvements

The RADIAN system will be battery-powered, providing wireless capabilities and flexible placement without requiring a constant wired connection. The radar subsystem will monitor human activity without capturing images, which allows for the protection of the privacy of users while still detecting falls. The AI model constantly monitors the radar subsystem's data to detect falls and then sends notifications. Alerts will be sent directly to the caregivers or emergency responders, allowing for a quick response time to accidents. The RADIAN system will operate autonomously, allowing users to have minimal interaction with the device.

5.2. Disadvantages and Limitations

Due to the limited power, since it will be battery-powered, this can restrict our complexity for our AI algorithm, which could lead to false alarms and/or missing scenarios that should trigger the alarm. Depending on the mmWave radar settings and wireless communication frequency, this can cause the battery life to be significantly reduced. In complex environments (cluttered rooms, multiple people, animals, children), the RADIAN system could produce more false positives. Depending on the size of the room, the RADIAN systems may not be able to detect falls in the outer areas of a large room. For the RADIAN system to effectively communicate with emergency services, it will need to depend on reliable network connectivity.

5.3. Alternatives

PIR Motion Sensors are used for simple movement/presence detection, are cheap and easy to use, but can not detect falls. Camera-Based Monitoring can provide a visual view of the room and verify if a fall occurs, offers more data for an AI model and could be used to create a better model without the need for complex data or high-resolution cameras, but raises privacy concerns for bedrooms and living spaces and requires high processing power, and storage could be an issue, leading to a high price point. A wearable device is capable of detecting a fall with an onboard sensor and is more accurate with little complexity, but it requires the user to wear a device constantly, and it may be uncomfortable, or the user could forget to wear the device, making it not as reliable.

5.4. Impact

Environmental Impact: The RADIAN device has low energy consumption. Minimal waste since the device will be compacted.

Society Impact: Increases safety for the elderly by providing rapid fall detection without invading privacy. Reduces the response times for caregivers and/or emergency services. Enables the elderly to live independently without sacrificing safety.

Ethical concerns: Allows for privacy due to no audio or video recording

RADIAN - RADar-based Intelligent
Activity moNitoring
Matthew Blackwell
Mitch Hoffman
Wyatt Johnson

FUNCTIONAL SYSTEM REQUIREMENTS

FUNCTIONAL SYSTEM REQUIREMENTS FOR RADIAN

TEAM 60

APPROVED BY:

Matthew Blackwell Date

Prof. Jang Date

Roman Venegas Date

Change Record

Rev	Date	Originator	Approvals	Description
1	10/04/2025	Matthew Blackwell		Draft Release
2	12/05/2025	Wyatt Johnson		Document formatting

Table of Contents

1. Introduction	6
1.1. Purpose and Scope	6
1.2. Responsibility and Change Authority	6
2. Applicable and Reference Documents	8
2.1. Applicable Documents	8
2.2. Reference Documents	8
2.3. Order of Precedence	8
3. Requirements	9
3.1. System Definition	9
3.2. Characteristics	10
3.2.1. Functional / Performance Requirements	10
3.2.2. Physical Characteristics	11
3.2.3. Electrical Characteristics	12
3.2.4. Environmental Requirements	13
4. Support Requirements	14
4.1. Wi-Fi Connection	14
4.2. Wi-Fi or Cellular Data Connection	14
4.3. Battery Maintenance	14
Appendix A: Acronyms and Abbreviations	15

List of Tables

Table 1: Subsystem Responsibility	7
Table 2: Applicable Documents for the RADIAN System	7
Table 3: Reference Documents for the RADIAN System	7

List of Figures

Figure 1: RADIAN Conceptual Image	1
Figure 2. Block Diagram of Radian System	4

1. Introduction

1.1. Purpose and Scope

Falls are a leading cause of injury among elderly individuals, often resulting in severe health consequences. Existing monitoring solutions on the market, such as wearable devices and camera-based systems, suffer from significant drawbacks. Wearable devices rely on user compliance, and camera-based systems compromise privacy. The RADIAN system will provide a low-cost, non-invasive, privacy-preserving fall detection system. The RADIAN system will use mmWave radar and AI algorithm technology to monitor human activity and detect falls in indoor environments, such as bedrooms and living rooms.

The RADIAN system will continuously monitor a room for motion patterns and classify them into fall/non-fall events in real time. Once a fall is detected, the RADIAN system will trigger an audible alarm and send a wireless alert to a caregiver's smartphone. Figure 1 shows the RADIAN system that is proposed in CONOPS.

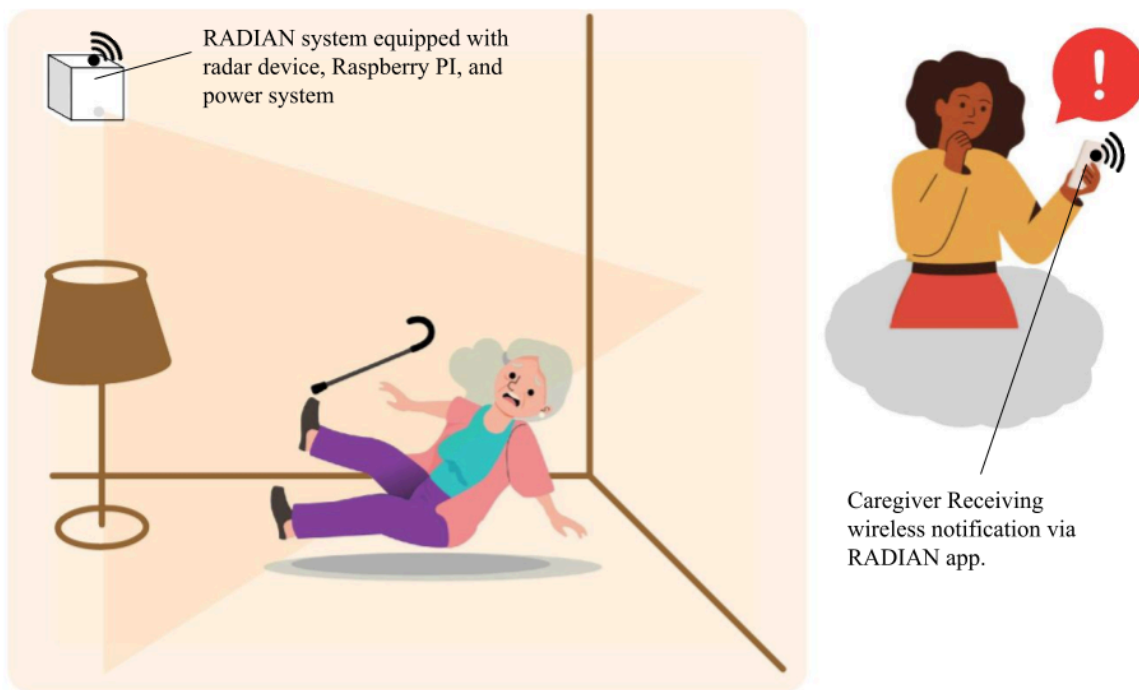


Figure 1: RADIAN Conceptual Image

This document describes detailed requirements for the RADIAN system, including the requirements for each subsystem (Physical, Wireless, and Data). The verification process for the RADIAN system will be described in a separate Validation Plan.

1.2. Responsibility and Change Authority

Matthew Blackwell, RADIAN's team leader, will be responsible for ensuring that all requirements are met for the RADIAN system. These requirements can only be changed

with approval from Matthew Blackwell and Professor Arya Menon, the RADIAN faculty Contact.

Subsystem	Responsibility
AI Model and Communication	Matthew Blackwell
Radar Sensing and App Development	Mitchell Hoffman
Power System and Physical Enclosure	Wyatt Johnson

Table 1: Subsystem responsibility

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

Document Number	Revision/Release Date	Document Title
IPC/WHMA-A-620E	January 2022	Requirements and Acceptance for Cable and Wire Harness Assemblies
UL 1977	December 7, 2022	Component Connectors for Use in Data, Signal, Control, and Power Applications

Table 2: Applicable Documents for the RADIAN System

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

Document Number	Revision/Release Date	Document Title
IEEE 802.11	Jan. 23, 1996	IEEE Standard for Information Technology – Telecommunications and Information Exchange between Systems

Table 3: Reference Documents for the RADIAN System

2.3. Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings, or other documents that are invoked as “applicable” in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICD,s which have their relevant documents considered to be incorporated as cited.

3. Requirements

3.1. System Definition

The RADIAN system is an all-in-one, designed system to help protect the elderly and high-risk individuals with quick and accurate alerts when the user has fallen. This allows the caregiver, whether a health care professional or loved one, to feel comfort knowing that their loved one is safe. The RADIAN system will comprise three overarching subsystems, each with further design requirements that must be met for the RADIAN subsystem. These three subsystems include Physical, Data, and Wireless.

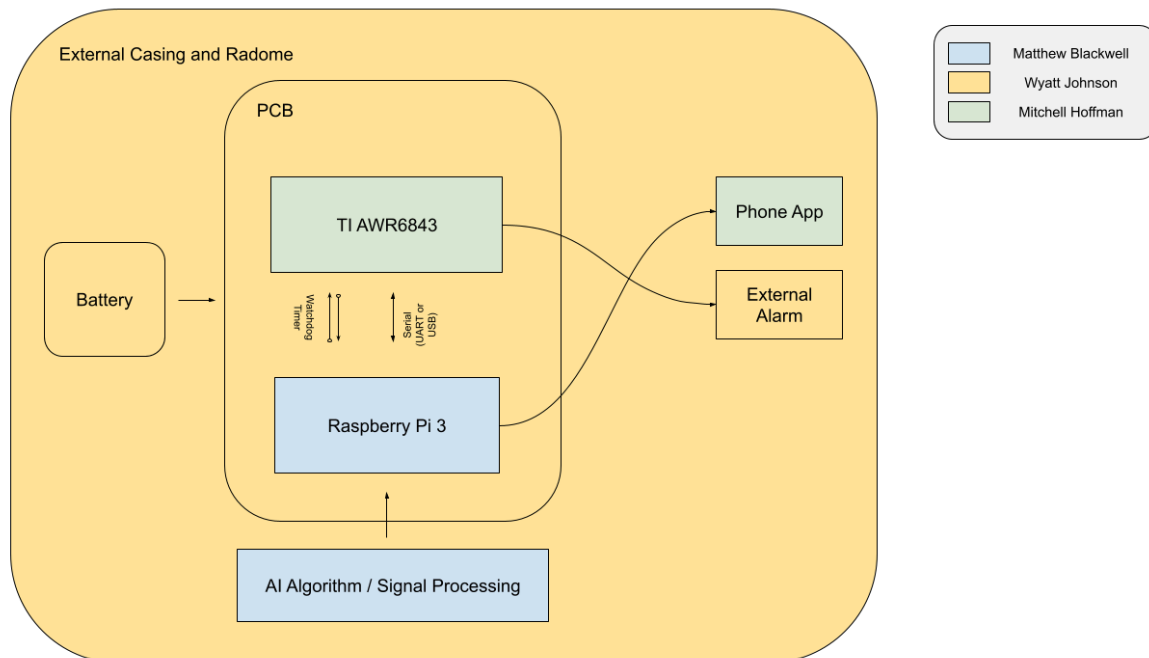


Figure 2. Block Diagram of Radian System

The entire RADIAN system is powered through two 3.7V rechargeable batteries, powering both the TI AWR6843ISK board and the Raspberry Pi 3, which are needed for the system to function. The rechargeable battery pack allows the user to easily charge the RADIAN system while maintaining maneuverability around the room. Both the TI board and Raspberry Pi 3 are connected to the custom-designed PCB, allowing for efficient connections and reduced risk of electrical connection failure between the subsystems. The ML model and AI algorithm are stored and run through the Raspberry Pi 3, where they will then communicate with an external phone application designed by the RADIAN group via a Gotify server. An external alarm will also be connected to the RADIAN system to alert anyone in the home while also alerting the caregiver's device.

The RADIAN system may have a feature that will trigger a sleep mode on the Raspberry Pi 3 to conserve power when no motion is detected in the room. Once motion is detected of an elderly or high-risk individual, the radar will continuously scan the room, sending the

processed data to the Raspberry Pi 3. Through its AI algorithm and ML model, the Raspberry Pi 3 will be able to compute in real-time the position, velocity, acceleration, and other fall parameters of the individual. If the ML model decides that a true fall has occurred based on a variety of parameters and decisions, the Raspberry Pi will send a message. The Gotify server will receive this new message and create a WebSocket, allowing for the message to travel in real time to the phone application. The custom-designed application will then receive this message and send a push notification to the caregiver, alerting them of the fall.

By combining these three subsystems, RADIAN provides a portable, intelligent, and connected solution for fall detection, balancing energy efficiency with reliable caregiver communication.

3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Fall Detection Accuracy

The RADIAN system shall detect falls with a minimum of 90% accuracy under normal indoor conditions (room temperature, clear view of the falling subject, unobstructed view, within 5 meters).

Rationale: The system will need to have a high fall detection accuracy to verify its capabilities for detecting falls reliably, which will also increase user trust.

3.2.1.2. Detection latency

Once a fall is detected, the RADIAN system will send an alert and sound an alarm within five seconds of a fall.

Rationale: It is critical that caregivers or emergency responders act as soon as possible, reducing the chance of an injury worsening.

3.2.1.3. Battery Life

The RADIAN system shall operate for a minimum of 3 hours on battery power without recharging.

Rationale: This amount of time will allow the system to stay functioning while the second battery is charging.

3.2.1.4. Communication Reliability

The RADIAN system shall send an alert over the Gotify server with a minimum success rate of 95%.

Rationale: A reliable communication system will ensure that fall events reach caregivers promptly, allowing for fast response times.

3.2.1.5. Local Alert Functionality

The RADIAN system will be equipped with a local audible alert system that, upon a fall being detected, a minimum of a 70 dB alarm will be activated.

Rationale: Having the local alert system will provide an alert to the user and nearby individuals that a fall has occurred. This will also allow for locating the victim quickly.

3.2.1.6. Autonomous Operation

The RADIAN system will operate autonomously, allowing users to have minimal interaction with the device.

Rationale: Autonomy ensures there will be continuous monitoring and timely alerts without the user needing to worry about maintaining the device.

3.2.2. Physical Characteristics

3.2.2.1. Radar Scan Area

The RADIAN system will operate in a standard room of 20 to 30 square meters with an angle of 45 degrees facing down.

Rationale: The RADIAN system is intended to operate in any standard household room, including but not limited to dining rooms, living rooms, and bedrooms. However, the RADIAN system is not rated for the bathroom.

3.2.2.2. Mass

The mass of the RADIAN System shall be less than or equal to 0.8 kilograms.

Rationale: A lighter system will be able to be mounted on walls without requiring heavy-duty hardware and support.

3.2.2.3. Volume Envelope

The volume envelope of the RADIAN System shall be less than or equal to 5 inches in height, 8 inches in width, and 14 inches in length.

Rationale: A compact system will be able to be mounted on walls, ceiling, or furniture, allowing for portability & relocation.

3.2.2.4. Mounting

The mounting information will be available in the RADIAN group's ICD as well as their user guide.

Rationale: The RADIAN system must be placed at a certain angle and height to capture the entirety of the room it is placed in.

3.2.2.5. Installation

The instructions for installing the RADIAN system are available for reference in the provided user guide. The installation required by the user will be mounting and battery charging/installation, as the phone application will already be installed and set up.

Rationale: The ease of installation allows the user to mount the RADIAN system and use it, ensuring that anyone can use the system.

3.2.3. Electrical Characteristics

3.2.3.1. Inputs

- a. The presence or absence of any combination of the input signals in accordance with ICD specifications applied in any sequence shall not damage the RADIAN system, reduce its life expectancy, or cause any malfunction, either when the unit is powered or when it is not.
- b. No sequence of commands shall damage the RADIAN System, reduce its life expectancy, or cause any malfunction.

Rationale: By design, it should limit the chance of damage or malfunction by user/technician error.

3.2.3.1.1 Power Consumption

The maximum peak power of the system shall not exceed 18 watts.

Rationale: Lower power draw will allow the system to run longer, while a high wattage will shorten the runtime.

3.2.3.1.2 Input Voltage Level

The input voltage level for the RADIAN system shall be 5.0V.

Rationale: This is the required voltage for devices used in the RADIAN system

3.2.3.2. Outputs

3.2.3.2.1 Data Output

The RADIAN system shall include a phone application interface compatible with the data system.

Rationale: The RADIAN system will pass information, specifically if a fall was detected, to a phone app designed for the caregiver of the high-risk individual.

3.2.3.2.2 Diagnostic Output

The RADIAN system may include a diagnostic interface for errors that occur within the system and debugging techniques for the system.

Rationale: This will provide a way for the use of the phone app to locate the error and provide solutions for fixing the error.

3.2.3.2.3 Wireless Communication Output

The RADIAN system shall wirelessly communicate with the phone application to alert if a fall is detected, using the Gotify server.

Rationale: Wireless communication will allow the caregiver to be notified, even if they are not present in the home to hear the audible alarm.

3.2.3.2.4 Cloud Data Storage

The RADIAN system will include a cloud storage option to upload data 4 seconds before the fall, during the fall, and 10 seconds after the fall.

Rationale: Cloud storage allows for first responders to know exactly when a fall occurs and the extent of the severity of the fall.

3.2.3.3. Connectors

The RADIAN System shall follow UL 1977 (2022), Component Connectors for Use in Data, Signal, Control, and Power Applications.

Rationale: Following UL 1977 will ensure that we use safe techniques and correctly rated connectors for lower power systems.

3.2.3.4. Wiring

The RADIAN System shall follow the guidelines outlined in IPC/WHMA-A-620 Class 1 safety requirements.

Rationale: Class 1 establishes basic workmanship for general, low-voltage products.

3.2.4. Environmental Requirements

The RADIAN System shall be designed to withstand and operate in the environments and laboratory tests specified in the following section.

Rationale: The RADIAN system will need to function in an indoor setting similar to the common home, with constant pressure and slight variations in temperature

3.2.4.1. Thermal

The RADIAN System will be able to operate in an environment between 50 °F and 80 °F

Rationale: The average temperature of many homes and buildings is between 50 °F and 80 °F, so the system must accommodate these varying temperatures

4. Support Requirements

4.1. Wi-Fi Connection

The RADIAN system requires a Wi-Fi connection to communicate with the Gotify server to send the REST POST notification. The user will provide the Wi-Fi connection to support the wireless connection of the RADIAN system.

Rationale: The system will need a Wi-Fi connection for the Raspberry Pi 3 to communicate with the Gotify cloud server. The Raspberry Pi 3's Wi-Fi connection can be changed remotely without having to view the GUI of the device.

4.2. Wi-Fi or Cellular Data Connection

The caregiver will be required to have a Wi-Fi or Cellular Data connection to receive the push notification from the RADIAN phone application.

Rationale: This allows the push notification to be communicated wirelessly over any distance.

4.3. Battery Maintenance

The user or caregiver must maintain the RADIAN system by occasionally changing the rechargeable batteries. They must also charge the depleted batteries to ensure there is always a backup for the system.

Rationale: Having battery maintenance allows the system to operate without protruding cables and will work if a power outage occurs.

4.4. Requirements for Customer

The RADIAN system requires a Wi-Fi connection to post an alert to the Gotify cloud server properly. Due to the system's nature, the caretaker will also need to download the phone application to receive push notifications indicating a fall has occurred. The user will have to provide the Wi-Fi connection, but this should not be an issue, as the majority of homes and buildings have some sort of Wi-Fi connection. The caretaker must have a wireless connection, either Wi-Fi or Cellular Data, to receive the push notifications.

Appendix A: Acronyms and Abbreviations

GUI	Graphical User Interface
RADIAN	RADar Intelligent Activity moNitoring

RADIAN - RADar-based Intelligent
Activity moNitoring
Matthew Blackwell
Mitch Hoffman
Wyatt Johnson

INTERFACE CONTROL DOCUMENT

INTERFACE CONTROL DOCUMENT FOR RADIAN

TEAM 60

APPROVED BY:

Matthew Blackwell Date

Prof. Jang Date

Roman Venegas Date

Change Record

Rev	Date	Originator	Approvals	Description
1	10/04/2025	Matthew Blackwell		Draft Release
2	12/05/2025	Wyatt Johnson		Document formatting and updated Electrical Power System

Table of Contents

Table of Contents	3
List of Tables	5
List of Figures	6
1. Overview	7
2. References and Definitions	8
2.1. References	8
2.2. Definitions	8
3. Physical Interface	9
3.1. Weight	9
3.2. Dimensions	9
3.2.1. Dimensions of Data Subsystem	9
3.2.2. Dimensions of Physical Subsystem	9
3.2.3. Dimensions of Wireless Subsystem	9
3.3. Mounting	9
3.3.1. Height	9
3.3.2. Angle	9
3.3.3. Locations	10
4. Thermal Interface	11
5. Electrical Interface	12
5.1. Primary Input Power	12
5.1.1 Electrical	12
5.1.2 Connector	12
5.1.3 Wiring	12
5.2. Charging Interface	12
5.2.1 Input	12
5.2.2 Connectors	12
5.3. Grounding and ESD	13
5.3.1 Ground	13
5.3.2 ESD	13
5.4. User Control Interface	13
6. Communications / Device Interface Protocols	14
6.1. Wi-Fi Communication	14
6.2. Cellular Data	14
6.3. Internet Protocols	14
6.4. UART Serial Interface	14
7. Mobile Application Interface	15
7.1 Communication Flow	15
7.2 User Interface	15
	3

7.3 Platform Compatibility	15
7.4 Notification Delivery	15
7.5 Performance Requirements	15

List of Tables

Table 1: Data Subsystem Dimensions	8
Table 2: Physical Subsystem Dimensions	8
Table 3: Wireless Subsystem Dimensions	8

List of Figures

Figure 1: Electrical Power System

9

1. Overview

This Interface Control Document (ICD) will cover the inputs, outputs, controls, and connections between the Physical, Data, and Wireless subsystems of the RADIAN system. This document will be continuously reviewed and updated as the RADIAN group continues to build, test, and integrate each subsystem. The ICD will include descriptions of each component being used in the system, how they interact with one another, and what data is output from each. The electrical connections and wireless interfaces, and protocols will be described in detail to ensure correct integration for success as described in the Concept of Operations (ConOps).

2. References and Definitions

2.1. References

IEEE 802.11

IEEE Standard for Information Technology – Telecommunications and Information Exchange between Systems

Jan. 23, 1996

2.2. Definitions

3GPP	3rd Generation Partnership Project
API	Application Programming Interface
A	Ampere
AI	Artificial Intelligence
CPU	Central Processing Unit
ConOps	Concept of Operations
g	Grams
GUI	Graphical User Interface
FSR	Functional System Requirements
HTTP	Hypertext Transfer Protocol
ICD	Interface Control Document
IEEE	Institute of Electrical and Electronics Engineers
ML	Machine Learning
mm	Millimeter
REST	Representational State Transfer

3. Physical Interface

3.1. Weight

The total weight of the RADIAN system, not including the Android device, will be less than or equal to 0.8 kilograms. This weight requirement enables the system to be properly mounted on the chosen room's wall without requiring heavy-duty hardware and support. Given that each subsystem has only a few components, the weight should not exceed the RADIAN group's given design.

3.2. Dimensions

3.2.1. Dimensions of Data Subsystem

Component	Length	Width	Height	Weight
Raspberry Pi 3	85.6 mm	56.5 mm	17 mm	41.2 g

Table 1: Data Subsystem Dimensions

3.2.2. Dimensions of Physical Subsystem

Component	Length	Width	Height	Weight
6.8Ah Power Board	135.38 mm	90.46 mm	<2 mm	<90 g
Radome	190 mm	120 mm	55 mm	<230 g

Table 2: Physical Subsystem Dimensions

3.2.3. Dimensions of Wireless Subsystem

Component	Length	Width	Height	Weight
TI AWR6843ISK	103 mm	57 mm	16 mm	91 g

Table 3: Wireless Subsystem Dimensions

3.3. Mounting

3.3.1. Height

The RADIAN system will need to be mounted at a minimum of 2 meters off the ground. This allows the system to properly scan the room, have good line-of-sight, and detect if a fall has occurred.

3.3.2. Angle

The RADIAN system must be mounted to the wall at a 45-degree angle to allow for the maximum amount of area to be scanned by the radar. Having the system parallel to the

ground may prevent certain areas of the room from being scanned and, therefore, prevent proper detection.

3.3.3. Locations

The RADIAN system can be located in any common room of a home or building, including but not limited to living rooms, dining rooms, bedrooms, and kitchens. The RADIAN system is not rated for humidity or water exposure, so it cannot be used in the bathroom.

4. Thermal Interface

With the Raspberry Pi 3 and TI AWR6843 chip being enclosed in a 3D printed structure, the added thermal heat could cause reliability issues as the RADIANT system runs. Because the system runs continuously, this further increases the chance for potential failures due to overheating. To mitigate these risks, a small heatsink will be added to the Raspberry Pi 3 to ensure that it remains cool as it handles the data and accurately predicts if a fall occurred. A small 30 mm fan will also be installed to provide airflow across both the Raspberry Pi 3 and the AWR6843 chip. These solutions will allow the RADIANT system to run reliably and efficiently.

5. Electrical Interface

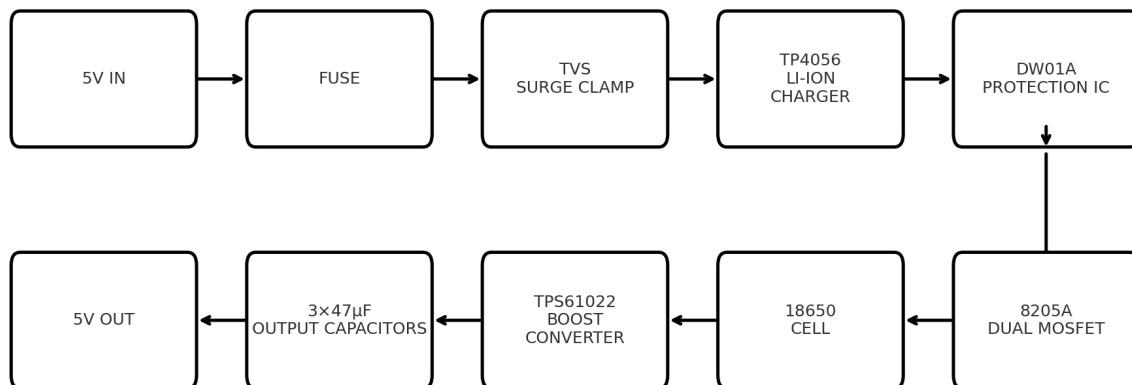


Figure 1: Electrical Power System

5.1. Primary Input Power

5.1.1 Electrical

The primary input power of the system will be two 18650 Li-ion cells in parallel, 3.7 V nominal each, for a 7 Ah total capacity. The battery voltage will be stepped up to 5V using a boost converter with a maximum continuous current of 2-3 A.

5.1.2 Connector

4-pin locking connector with two positive terminals and two negative terminals for the Raspberry Pi 3 and TI AWR6843.

Pinout (example): 1 = BAT+, 2 = BAT+, 3 = GND, 4 = GND.

5.1.3 Wiring

18–20 AWG wiring between the power board and the Raspberry Pi 3/ TI AWR6843. (Target $\leq 2\%$ drop on the 5 V rail.)

5.2. Charging Interface

5.2.1 Input

The battery system will be charged using a 5V, 1A wall adapter barrel jack. Charging at a low current will keep the batteries from overheating and ensure safety.

5.2.2 Connectors

The battery system will receive its power through a 5V barrel jack connected to a wall adapter.

5.3. Grounding and ESD

5.3.1 Ground

Single ground plane located on the power PCB.

5.3.2 ESD

TVS diodes on the barrel jack and on external connectors.

5.4. User Control Interface

The RADIAN system will come with a custom-designed phone application where the caregiver will be able to receive push notifications that a fall has been detected. The app may include a simple battery monitoring system to alert the user when the batteries need to be recharged.

6. Communications / Device Interface Protocols

6.1. *Wi-Fi Communication*

The Raspberry Pi 3 will use the IEEE 802.11b/g/n Wi-Fi standards through its onboard communication chip. These standards apply to Wi-Fi 1, Wi-Fi 3, and Wi-Fi 4, each offering new advancements in performance.

6.2. *Cellular Data*

The Gotify server will communicate with the phone application by 5G cellular data, given by the 3GPP standards for 5G wireless communication.

6.3. *Internet Protocols*

The Raspberry Pi 3 will post the alert message through the standard HTTP REST API protocol, allowing the message to be communicated from the Raspberry Pi 3 to the Gotify server.

6.4. *UART Serial Interface*

A UART serial interface will connect the Raspberry Pi 3 to the TI AWR6843ISK. Data shall be exchanged using packets containing radar object data. The Raspberry Pi shall use these packets and convert the radar output into the AI algorithm, which is then sent to the Gotify server.

7. Mobile Application Interface

7.1 Communication Flow

The Raspberry Pi 3 shall send event data to the Gotify server using an HTTP POST request. The Gotify server will then relay the data to the mobile app via WebSocket connection to receive messages in real time.

7.2 User Interface

The app shall display incoming alerts and system messages, battery life, and previous alerts, if any, in a custom-designed user interface.

7.3 Platform Compatibility

The app will be developed using Flutter for the user interface and Dart for the programming language. This gives the ability for the app to be run on Android devices.

7.4 Notification Delivery

The app shall generate a push notification when it has received an alert message from the Gotify server. The notifications will be displayed using the notification system of the Android device, to ensure the notifications are visible when the app is not open.

7.5 Performance Requirements

The alerts shall be delivered to the app within 5 seconds of the Raspberry Pi sending the alert to the Gotify server, under normal network conditions.

RADIAN - RADar-based Intelligent Activity moNitoring

Matthew Blackwell

Mitch Hoffman

Wyatt Johnson

EXECUTION TIMELINE

[illegible]

RADIAN - RADar-based Intelligent Activity moNitoring

Matthew Blackwell

Mitch Hoffman

Wyatt Johnson

VALIDATION

Power Subsystem			
Test	Description	System	Status
WEBENCH Design Simulation	Steady-state, startup, load-transient, and input-transient simulations validate TPS61022 component choices and operating margins	Boost	Completed
ERC/DRC Sign-off	Run schematic/PCB ERC/DRC and waive labeling issue on charger/boost connection	Power Board	Completed
Footprint/BOM Cross-Check	Verify charger, boost, holders, and connector footprints and pin mappings against datasheets	Power Board	Completed
Power Tree Review	Review voltage and current budgets against loads and IC ratings.	Power Board	Completed
Visual & Dimensional Check	Inspect bare PCB size, hole locations, silkscreen, and solder mask vs. the fab drawing	Power Board	Completed
Solderability Check	Test pad finish quality by tinning a coupon/pad and inspecting wetting	Power Board	Completed
Fit Check: 18650 Holders & Connectors	Verify 18650 holders, barrel jack, and connectors seat flat and align with holes before reflow	Power Board	Completed
Reflow Profile Check	Apply paste and reflow; confirm thermal profile is within datasheet limits for ICs	Assembly	Completed
Microscope Inspection	Inspect IC pins and passives for bridges, voids, tombstones, and poor fillets	Assembly	Completed
Polarity/Orientation	Check diodes, LEDs, electrolytics, and IC pin 1 orientation vs. silkscreen	Assembly	Completed
Resistance-to-GND Check	Measure resistance from +BAT and +5 V rails to GND to catch gross shorts before power-on	Power Board	Completed
Limited Bench Power-On	Power VIN from bench supply with current limit; confirm reasonable idle current and no hot spots	Charger/ Boost	Completed

Key Node DC Sanity	Check VIN, battery node, and 5 V rail voltages with no load	Power Board	Completed
Charger Bench Emulation	Emulate battery with bench supply; verify TP4056 charge behavior and status pins vs. datasheet	Charger	Completed
Single-Cell Charge Profile	Charge 18650 cell; log battery voltage over time to confirm CC/CV behavior and termination	Charger	Completed
Dual-Cell Pack Charge Curve	Charge 2 * 18650 batteries; record battery voltage vs. time	Charger	Completed
Charge + Boost Operation	Run TP4056 charging while the boost converter supplies 5 V rail; check for heating or instability	System	Completed
No-Load Output	Measure 5 V rail at no load; verify approximately 5.0 V regulation	Boost	Completed
Startup Under Load	Verify the boost circuit can start up under load	Boost	Completed
Raspberry Pi Load Test	Power Raspberry Pi 3 from the board; verify successful boot and stable ~4.9–5.0 V rail	RADIAN System	Completed
Audible Alarm	Test the decibel of the audible alarm	RADIAN System	Future work

Radar Subsystem			
Test	Description	System	Status
UART Communication Integrity Test	Radar correctly transmits binary TLV packets and Python script decodes them	Radar	Completed
Point Cloud Accuracy Test (static target)	Radar accurately reports x, y, z coordinates of stationary object	Radar	Completed
Point Cloud Accuracy Test (moving target/velocity)	Radar accurately reports Doppler-based velocity estimation	Radar	Completed
Multiple Object Detection	Radar can detect more than one target simultaneously	Radar	Completed

Test			
Configure File Test (.cfg integration)	Ensure custom configurations change radar behavior	Radar	Completed
Real Time Terminal Display Test	Python file displays live point cloud data without delay or crash	Radar	Completed

App Subsystem			
Test	Description	System	Status
App Navigation Test	All tabs function and route correctly	App	Completed
Alert Simulation Test	Fall simulation button triggers critical alert logic	App	Completed
Settings and Server Connection Test	Validate server connection workflow	App	Completed
Gotify Push Notification Test	App receives live notifications from server	App	Completed
Weekly Fall Log Generation Test	Validate automatic weekly log creation	App	Completed
Data Persistence Test	Fall logs persist after app close/restart	App	Completed
UI Responsiveness Test	App layout works on multiple screen sizes	App	Completed

Notification Subsystem			
Test	Description	System	Status
Pi-Server Connectivity Test	Verify Raspberry Pi can reach the laptop/host running Gotify via local IP (ping test)	Notification Server	Completed
Gotify Installation	Confirm Gotify server is installed on the Raspberry Pi by inspecting	Notification	Completed

Verification	the pip list and verifying the package	Server	
Gotify UI and Token Setup	Test Gotify UI in a web browser via the laptop IP address and create/share the application token for RADIANT	Notification Server	Completed
Basic Message Publishing	Test Python script sending a “Hello World” message via HTTP POST from the Pi to Gotify and verify it appears in the Gotify UI	Notification Server	Completed
Invalid Token Handling	Test server behavior with an incorrect app token and confirm that an error is reported, and the message is not accepted	Notification Server	Completed
Repeated Message Stability	Test repeated message sends in rapid succession from Pi to Gotify and verify no alerts are dropped	Notification Server	Completed
Server Restart Recovery	Restart the Gotify server and confirm it resumes normal operation and can receive new messages	Notification Server	Completed

Artificial Intelligence Subsystem			
Test	Description	System	Status
Model Training & Confusion Matrix Check	Verify the random forest classifier correctly separates static vs dynamic motion; inspect accuracy and confusion matrix on synthetic data	Artificial Intelligence	Completed
Synthetic Frame Verification	Run the classifier on known synthetic test frames and verify that predicted labels match the expected labels	Artificial Intelligence	Completed
Edgecase Generation & Inclusion	Generate edge cases (sitting, bending, kneeling, lying down slowly, trip recovery, crawling, pet walking, object drop, hard fall) and include them in training	Artificial Intelligence	Completed
Fall Logic Edgecase Validation	Run the full pipeline (preprocessing, classification, fall detection) on edge cases to verify “No Fall Detected” in non-fall scenarios	Artificial Intelligence	Completed

Normal Fall Scenario Validation	Run the full pipeline on normal fall test cases to verify the system outputs “Fall Detected” when appropriate	Artificial Intelligence	Completed
Demo Accuracy Verification	Confirm that the ML model reaches the current 89% accuracy target during demo conditions, meeting the verification plan	Artificial Intelligence	Completed
Real-Time Performance on Raspberry Pi	Check that point-cloud processing, ML inference, and fall logic run within real-time constraints on the Raspberry Pi without overloading the system	Artificial Intelligence	Completed

RADIAN - RADar-based Intelligent
Activity moNitoring
Matthew Blackwell
Mitch Hoffman
Wyatt Johnson

SUBSYSTEM REPORTS

SUBSYSTEMS REPORT FOR RADIAN

TEAM 60

APPROVED BY:

Matthew Blackwell Date

Prof. Jang Date

Roman Venegas Date

Change Record

Rev	Date	Originator	Approvals	Description
1	11/28/2025	Matthew Blackwell		Original Release

Table of Contents

Table of Contents	3
List of Tables	5
List of Figures	6
1. Introduction	7
2. Power Subsystem Report	8
2.1. Subsystem Introduction	8
2.2. Subsystem Details	8
2.2.1. Batteries and Storage	8
2.2.2. Charging	8
2.2.3. 5 V Conversion and Distribution	9
2.2.4. Loads and Estimated Current Draw	9
2.3. Subsystem Validation	10
2.3.1 Boost Converter Design Simulations	10
2.3.2 Benchtop Measurements	12
2.4. Subsystem Conclusion	13
3. mmWave Radar Subsystem Report	14
3.1 Subsystem Introduction	14
3.2 Subsystem Details	14
3.2.1 Hardware Platform and Radar Configuration	14
3.2.2 UART Communication Architecture	14
3.2.3 TLV Frame Parsing and Point Cloud Extraction	14
3.2.4 Integration With the Data Subsystem	14
3.3 Subsystem Validation	15
3.3.1 Functional Testing With Live Radar Data	15
3.3.2 Configuration File Testing	15
3.3.3 TLV Parsing Verification	15
3.3.4 Edge Case Testing	15
3.4 Subsystem Conclusion	15
4. Mobile Application Subsystem Report	16
4.1 Subsystem Introduction	16
4.2 Subsystem Details	16
4.2.1 Application Architecture	16
4.2.2 Gotify Integration and Real Time Alerts	16
4.2.3 Fall Log Feature and Local Database	16
4.2.4 Theming and User Interface Experience	17
4.3 Subsystem Validation	17
4.3.1 UI and Workflow Testing	17
4.3.2 Alert Handling and Simulation Testing	17

4.3.3 Database Structure Verification	17
4.3.4 Stability and Performance Testing	17
4.4 Subsystem Conclusion	17
5. Notification Server Subsystem Report	18
5.1 Subsystem Introduction	18
5.2 Subsystem Details	18
5.2.1 System Architecture	18
5.2.2 Server Platform	18
5.2.3 Raspberry Pi Integration	18
5.3 Subsystem Validation	19
5.3.1 Server Deployment and Connectivity Testing	19
5.3.2 Message Publishing and Payload Validation	19
5.3.3 Stability and Performance Testing	19
5.4. Subsystem Conclusion	19
6. Artificial Intelligence Subsystem Report	20
6.1 Subsystem Introduction	20
6.2 Subsystem Details	20
6.2.1 System Architecture	20
6.2.2 Machine Learning Model Design	20
6.2.3 Fall Detection Algorithm	21
6.2.4 Raspberry Pi Integration	21
6.3 Subsystem Validation	21
6.3.1 Model Training and Functional Verification	21
6.3.2 Edgecase Generation	22
6.3.3 Frame Classification and Fall Logic Validation	22
6.3.4 Stability and Performance Testing	22
6.4. Subsystem Conclusion	22

List of Tables

Table 1: Estimated Current Draw of RADIAN System.....7

List of Figures

Figure 1: Functional block diagram of the power subsystem	8
Figure 2: steady-state output voltage of the TPS61022 at nominal load	10
Figure 3: TPS61022 Vout vs Time and Vin vs Time.....	11
Figure 4: TPS61022 Vout vs Time and Iout vs Time during load transient simulation.....	11
Figure 5: TPS61022 Vout vs Time and Iout vs Time during input-voltage transient simulation.....	12
Figure 6: Voltage of batteries vs Time (s) while charging.....	13

1. Introduction

The RADIAN system is a radar-based fall-detection device that is designed to autonomously monitor activity in indoor living spaces and then notify caregivers when a high-risk individual person falls. The system collects data via a mmWave radar and uses a machine-learning algorithm to determine if a fall has occurred. After a fall has occurred, the Raspberry Pi will deliver an alert to a mobile application through a notification server. RADIAN is organized into the Power, mmWave Radar, Artificial Intelligence (Data), Notification Server, and Mobile Application subsystems, each with defined roles and interfaces. Every subsystem was validated to meet its requirements. Through this validation, a clear path to full integration into the complete system defined in the ConOps, FSR, and ICD documents will be possible.

2. Power Subsystem Report

2.1. Subsystem Introduction

The RADIANT power subsystem is designed to provide a stable 5 V supply to the Raspberry Pi 3 and the TI AWR6843ISK mmWave radar evaluation board. The subsystem will utilize 18650 rechargeable Li-ion batteries that will be recharged via a 5 V barrel jack. This will allow the system to be placed in any location and not depend on wall outlets for power.

The goals of the power subsystem are to deliver a regulated 5 V rail with sufficient current for the Raspberry Pi and radar. Safely charge and protect the Li-ion battery pack. Protect the rest of the system from faults (over-voltage, short circuit, reverse polarity). Provide several hours of runtime from the battery alone.

The power subsystem was tested to confirm its stability, load capability, and basic safety functions. These tests help verify that the subsystem will reliably support the Raspberry Pi and the mmWave radar of the RADIANT system.

2.2. Subsystem Details

A simplified block diagram of the subsystem is below.

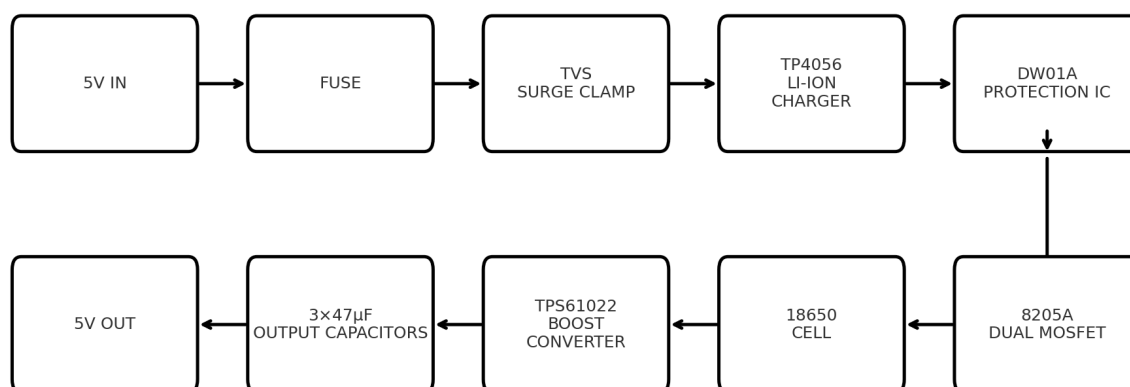


Figure 1: Functional block diagram of the power subsystem

2.2.1. Batteries and Storage

The power subsystem uses two 18650 Li-ion cells (nominal 3.7 V) in parallel to have a total capacity of 6800 mAh. A dedicated protection circuit (DW01A + dual MOSFET pack) provides over-charge, over-discharge, and over-current protection at the cell level. The total stored energy is based on the two 3400 mAh cells in parallel, which provide roughly 25.2 Wh of usable energy at 3.7 V nominal.

2.2.2. Charging

A single-cell Li-ion charger IC (TP4056) is used to charge the two batteries from a 5 V barrel jack input. The charger regulates the charge current to a safe level for the cells and will terminate the charge when the battery reaches 4.2V. The design supports up to 2 A input currents. The charger IC will charge the two batteries at a constant current of 1 A.

Although it is not recommended to charge and use the system due to inefficient charging of batteries, the system can operate while the batteries are charging.

2.2.3. 5 V Conversion and Distribution

A synchronous boost converter (TPS61022) steps the battery voltage from 3.0-4.2V up to a regulated 5 V system rail. The boost converter and board layout are designed for a constant output current of 2-3 A, with short-term peaks above the current limit for inrush or transient loads. For output filtering, there are 3, 47 μ F ceramic capacitors that help reduce ripple and prevent the 5 V rail from sagging during sudden load increase, such as Raspberry Pi boot or radar startup.

Before finalizing the boost converter design, the TPS61022 circuit was created and analyzed using TI's WEBENCH design tool. WEBENCH was used to select all the necessary components to allow for a 5V, 2-3 A output from a 3.0-4.2V Li-ion input. The simulations that were run provide an estimated efficiency, output ripple, and component stress, which were used to confirm that the boost design could supply the necessary load currents before building the hardware.

2.2.4. Loads and Estimated Current Draw

The primary 5 V loads of the RADIANT system are the Raspberry Pi 3, TI AWR6843ISK mmWave radar evaluation board, and charging status LEDs. The table below shows the approximate current draw for each component.

Component	Voltage	Typical Current Draw	Notes
Raspberry Pi 3	5 V	0.35-0.95 A	0.35 A idle, up to 0.95 A under 400% CPU load
AWR6843ISK mmWave Radar	5 V	0.4-0.8 A (estimated)	Typical current from documentation
Status LEDs	5 V	<10mA	Calculated based on the series resistor

Table 1: Estimated Current Draw of RADIANT System

Under normal operation of the Raspberry Pi and TI mmWave radar, the total average current is around 1.0-1.8 A, with short peaks on start-up or heavy computations. The boost converter, layout, and connector sizing were selected to comfortably support these loads.

Using the 25.2 Wh battery pack calculation and assuming around 85-90 % efficiency in the boost converter, a continuous 5 V load of roughly 1.5 A (7.5W) corresponds to a runtime of 2.8-3 hours. Actual runtime will vary depending on how heavily the Raspberry Pi and radar are used.

2.3. Subsystem Validation

The RADIANT power subsystem was validated through a series of simulations and benchtop tests to validate the boost converter, charger, and protection circuits, as well as to verify that the system can reliably power the Raspberry Pi and radar.

2.3.1 Boost Converter Design Simulations

Prior to building the hardware, the boost converter design was validated using various simulations in TI's WEBENCH design reports.

2.3.1.1 Steady-state analysis

This simulation verified that the TPS61022 was capable of maintaining a regulated 5 V output over the expected input range and load currents while also staying within an acceptable output ripple and inductor/capacitor stress limits.

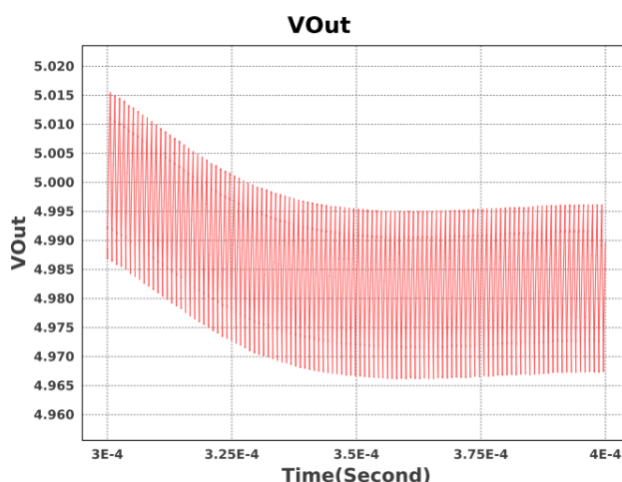


Figure 2: steady-state output voltage of the TPS61022 at nominal load

Figure 2 shows the 5 V output voltage versus time at nominal load. The x-axis is time (seconds), and the vertical axis is VOUT (volts). The red band represents the output ripple around 5 V. We can see that VOUT remains between 4.8 V and 5.01 V, showing a stable, well-regulated output.

2.3.1.2 Startup analysis

This simulation verified that the TPS61022 could start up cleanly under load without having excessive overshoot on the 5 V rail. This will help understand the boost converter capabilities when the Raspberry Pi and TI mmWave radar are connected to the power board.

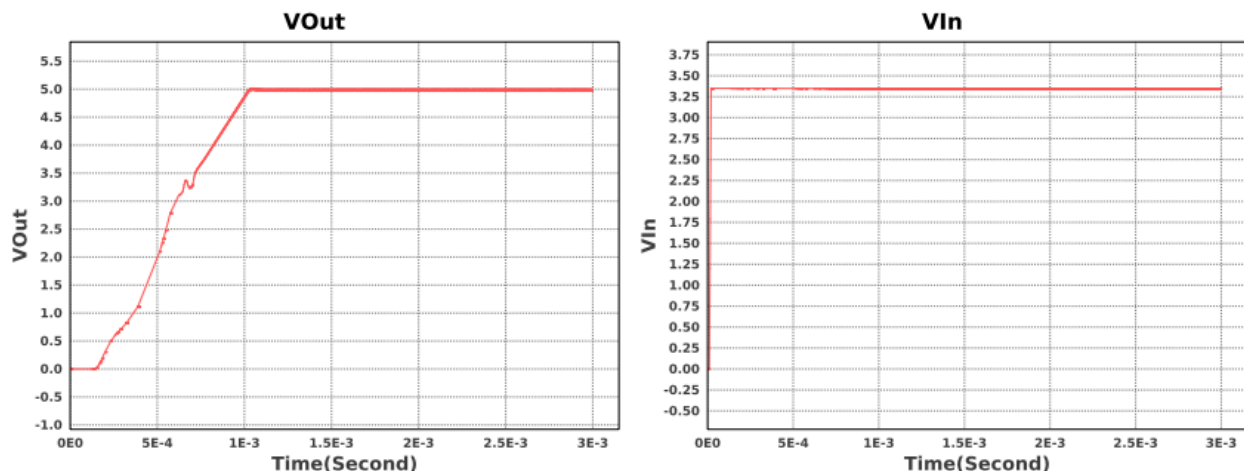


Figure 3: TPS61022 Vout vs Time and Vin vs Time

Figure 3 shows the simulated startup behavior of the TPS61022. The left graph shows VOUT rising from 0 V to ~5 V in about 1ms with almost no overshoot. The right graph shows VIN remaining nearly constant at about 3.4 V during this time. These figures confirm that the converter starts up quickly and smoothly with little to no voltage spikes on the 5 V rail.

2.3.1.3 Load-transient analysis

This simulation verified that the TPS61022 5 V output response to step changes in current, similar to the Raspberry Pi and TI mmWave radar switching between idle and active states.

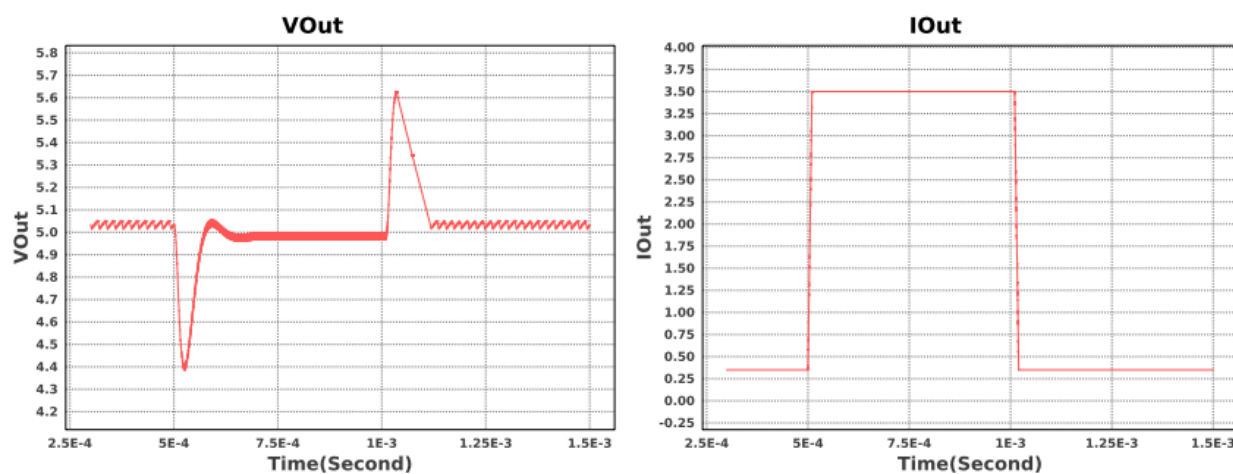


Figure 4: TPS61022 Vout vs Time and Iout vs Time during load transient simulation

Figure 4 (left graph) shows the simulated 5 V output voltage during a load step, and (right graph) shows the corresponding output current. The load current steps from ~0.3 A to ~3.5 A and then back down to ~0.3 A. This step causes a brief overshoot and undershoot on the 5 V rail. In both cases, the output quickly settles back to ~5 V, showing that the converter can handle rapid changes in load without large or long-lasting spikes on the supply.

2.3.1.4 Input-transient analysis

This simulation verified that the TPS61022 could handle the input voltage changing rapidly, emulating the changes in battery voltage, i.e., changing the battery cells.

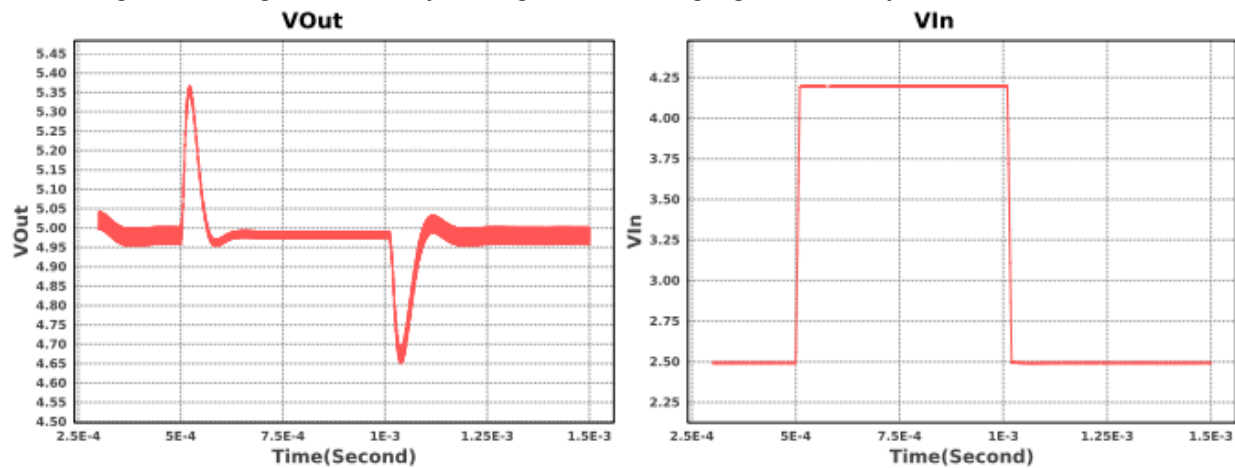


Figure 5: TPS61022 Vout vs Time and Iout vs Time during input-voltage transient simulation

Figure 5 (left graph) shows the simulated 5 V output voltage, and (right graph) shows the corresponding input voltage. In the simulation results, V_{in} steps from ~ 2.5 V to ~ 4.25 V and then back down to ~ 2.5 V. As V_{in} changes, V_{out} shows brief spikes before quickly returning to its steady 5 V output. This shows that the converter maintains regulation on the 5 V rail even when the battery voltage changes suddenly.

2.3.2 Benchtop Measurements

After validating the design, the RADIANT power subsystem was assembled and tested. These tests were used to verify that the hardware meets the design goals for output regulation, load capability, and safe battery charging.

2.3.2.1 Voltage and Current Testing

The first tests were performed with no load on the output. With no external load connected, the boost converter output measured approximately ~ 5.03 V.

During operation, an unknown amount of current caused a high-current thin trace between the inductor and the TPS61022 switching pin to overheat and open. This is not due to the TPS61022 limitations, but the copper trace design being too thin (5 mil). The failure was noted, and future revisions will increase the trace width to increase current limits. Due to this flaw, load testing was not extensively tested to ensure a fully functional board before the demo.

2.3.2.2 Raspberry Pi load test

After completing resistive load testing, the power board was connected directly to a Raspberry Pi 3 to confirm that the subsystem was capable of powering a real digital load. The Pi booted while the 5 V rail was measured using a digital multimeter. During the boot, the measured voltage remained at a constant ~ 4.92 V. This is well within the range that the recommended voltage stays at for the Raspberry Pi.

2.3.2.3 mmWave Radar load test

The TI AWR6843ISK mmWave radar was not directly connected to the power board during testing. Because this board is shared with another member's subsystem and there was limited time remaining before the final demo, we chose not to risk damaging the radar board.

2.3.2.4 Charger and battery behavior

The TP4056 charging circuit was tested by connecting two semi-discharged 18650 batteries in parallel to the board and applying a 5 V input at the barrel jack. The charge-status LED indicators on the TP4056 behaved as expected (red LED on when charging, green LED on when charging is complete). Below are the voltage results of the two batteries recorded over 4 hours and 44 minutes.

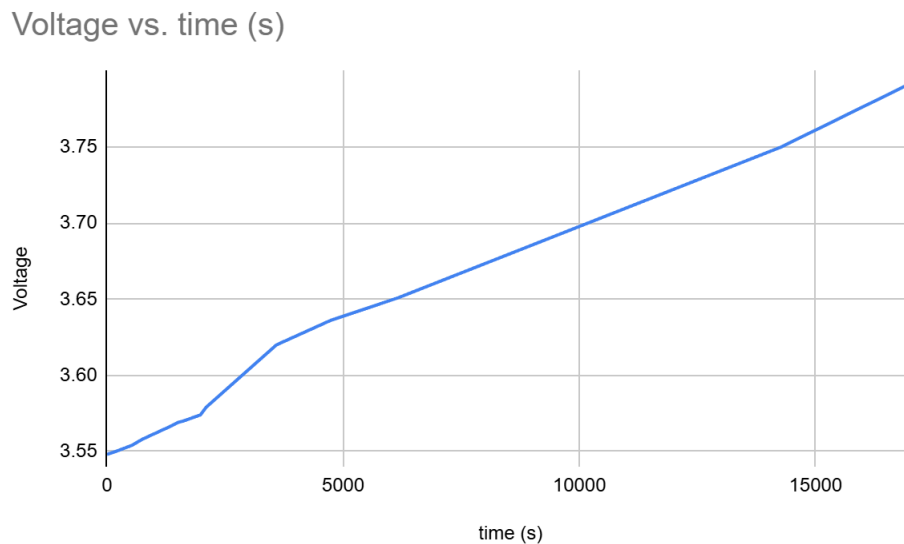


Figure 6: Voltage of batteries vs Time (s) while charging

Analyzing **Figure 6**, we can see that the 18650 batteries are reliably charging. While the charge time is long, this is expected since the battery's total capacity is 6.8Ah.

During charging, the boost circuit output was measured at ~5 V while the batteries were charging. Even though this mode is less efficient at charging the batteries and is not recommended for system use, no heating or instability issues were observed during these tests.

2.3.2.5 Audible alarm

In the FSR and ConOps, a local audible alarm is planned but not yet implemented. This will be implemented from January to May.

2.4. Subsystem Conclusion

Overall, the WEBENCH simulations and physical measurements show that the RADIAN power subsystem meets the core 5 V regulation and Pi load requirements. The boost converter maintains stability across the expected range, the charger safely charges the Li-ion batteries, and the hardware can power a Raspberry Pi 3 reliably. This validated the

performance of the power subsystem and provided confidence that the RADIAN system will have a reliable and stable power supply, while also providing valuable layout and protection improvements for future revisions.

3. mmWave Radar Subsystem Report

3.1 Subsystem Introduction

The mmWave Radar Subsystem is responsible for detecting human presence, monitoring motion, and identifying potential fall events within the RADIAN system. This subsystem uses the Texas Instruments AWR6843ISK radar and a custom Python based UART processing pipeline to extract real time point cloud data. The subsystem serves as the primary sensing module for fall detection and provides the Raspberry Pi Data Subsystem with continuous spatial and velocity information needed for classification algorithms.

3.2 Subsystem Details

3.2.1 Hardware Platform and Radar Configuration

The subsystem uses the TI AWR6843ISK, a 60-64 GHz FMCW mmWave radar supporting range, Doppler (velocity), and angle estimation. A configuration file (.cfg) generated from TI's mmWave Visualizer defines the chirp parameters, sampling settings, and processing chains for indoor monitoring. The Python pipeline automatically transmits this configuration to the radar during initialization, ensuring the sensor operates in a mode suitable for fall detection.

3.2.2 UART Communication Architecture

The subsystem uses two UART channels, a command line interface UART operating at 115,200 baud for sending configuration commands and a high speed data UART operating at 921,600 baud for receiving binary radar output. The mmwave_run.py script establishes and manages both ports, transmits the full configuration file, and maintains continuous monitoring of incoming radar data. This arrangement enables the radar to receive initialization commands and deliver live measurement data without interruption.

3.2.3 TLV Frame Parsing and Point Cloud Extraction

The radar outputs its processed data in a binary TLV (Type-Length-Value) format. The Python pipeline reads these TLVs, identifies the "Detected Points" section, and converts the spherical radar measurements into x, y, and z coordinates, along with velocity values derived from Doppler frequency shifts. The script formats each frame as a separate JSON object using NDJSON formatting, which simplifies integration with downstream software by keeping the data stream clean and machine readable while ensuring debug output remains isolated on STDERR.

3.2.4 Integration With the Data Subsystem

The radar subsystem is designed to integrate with the Raspberry Pi Data Subsystem. Although it currently operates independently for testing and development, it will eventually run on the Raspberry Pi, continuously streaming JSON formatted point cloud data to the data processing modules. This real time stream will support AI based fall detection algorithms and enable the collection of presence, motion, and velocity information.

3.3 Subsystem Validation

3.3.1 Functional Testing With Live Radar Data

Validation involved operating the radar in indoor environments and confirming that the system correctly identified stationary users, tracked walking motion, and captured rapid movements associated with fall simulations. Real time output in the terminal verified that the TLV frames were being interpreted correctly and that the point cloud data remained stable and continuous.

3.3.2 Configuration File Testing

Multiple radar configuration profiles were evaluated to ensure the pipeline operated correctly across different modes. These included short range and medium range indoor profiles. Each configuration produced expected point cloud characteristics, such as consistent frame rates and appropriate numbers of detected points.

3.3.3 TLV Parsing Verification

TLV parsing was validated by monitoring the decoded header fields, TLV counts, and detected point quantities produced by the Python script during live testing. The reported packet lengths and number of TLVs consistently matched the expected structure defined by the TI AWR6843ISK data format. In addition, the point cloud output behaved as expected when a person stood or moved at known positions relative to the radar.

3.3.4 Edge Case Testing

The subsystem was tested under various conditions, including empty rooms, cluttered environments, multiple moving subjects, and different mounting angles for the radar. In each scenario, the pipeline maintained stable operation and produced valid JSON frames without crashes or distorted output.

3.4 Subsystem Conclusion

The mmWave Radar Subsystem successfully demonstrated reliable point cloud data, TLV frame parsing, and UART based communication using the AWR6843ISK platform. Testing confirmed that the radar pipeline consistently produced accurate spatial and velocity data suitable for fall detection algorithms, and the subsystem remained stable across various operating conditions. Although system level integration is still forthcoming, the radar subsystem has been fully developed and validated in isolation, confirming its readiness for incorporation into the complete RADIAN sensing pipeline.

4. Mobile Application Subsystem Report

4.1 Subsystem Introduction

The Mobile Application Subsystem provides caregivers an interface for the RADIAN system and plays a central role in delivering fall alerts, presenting system status, and maintaining historical event records. Developed using Flutter for Android devices, the application incorporates a modular, tab based architecture that separates alert handling, historical logging, settings configuration, and system monitoring into distinct, intuitive interfaces. The subsystem is designed to integrate with the RADIAN backend through a Gotify server, enabling real time delivery for fall notifications generated by the mmWave radar subsystem.

4.2 Subsystem Details

4.2.1 Application Architecture

The mobile application follows a tab based navigation structure that organizes the system's functionality across four main interfaces, the Home, Alerts, Fall Log, and Settings tabs. The Home tab displays overall system status and highlights active alerts. The Alerts tab allows users to simulate fall events and is designed to display real time alerts received from the Gotify notification server. The Fall Log tab provides caregivers with access to historical fall data and is backed by a local SQLite database implemented through Drift. This database stores fall events in a persistent on device file and supplies the Fall Log interface with structured records for review. The Settings tab enables configuration of the Gotify server endpoint and authentication token, and it also includes theme management options that let users choose between light mode or dark mode for the system's theme preferences. These components form a modular and scalable application architecture in which the user interface, local storage subsystem, and networked alerting interface are implemented independently but designed for seamless integration.

4.2.2 Gotify Integration and Real Time Alerts

The application was designed to receive real time alerts through a Gotify server, which serves as the communication bridge between the Raspberry Pi data subsystem and the mobile device. A custom Dart service manages server configuration, authentication token storage, message parsing, and incoming alert handling. When an alert is received, the app presents a notification and initiates a recurring reminder loop that generates additional notifications every thirty seconds until the alert is acknowledged by the caregiver.

4.2.3 Fall Log Feature and Local Database

Persistent event storage is implemented using Drift, which provides a SQLite database stored locally on the device. The database maintains fall event timestamps and event source information. The Fall Log tab queries this database and displays entries chronologically for caregiver review. Although live system alerts were not yet available, the database schema and retrieval methods were validated using simulated entries, ensuring the subsystem is ready for integration.

4.2.4 Theming and User Interface Experience

The application uses Flutter's Material 3 design system and a custom RADIAN color theme. Users may select light mode or dark mode for the app theme through controls provided in the Settings tab. UI elements were tested on multiple screen sizes to confirm consistent layout, visibility of alerts, and smooth navigation across tabs.

4.3 Subsystem Validation

4.3.1 UI and Workflow Testing

The user interface and navigation flow were validated using Android emulators to ensure that each tab behaved as intended. Simulated alerts were used to verify that the Alerts and Home tabs correctly displayed active notifications and that the acknowledgement workflows operated smoothly. Navigation between tabs, theme changes, and general responsiveness were also confirmed.

4.3.2 Alert Handling and Simulation Testing

Because there is not a system wide integration, alert validation focused on simulated fall events generated within the application. These simulations confirmed that the app could present fall alerts, highlight them in the UI, and clear them upon acknowledgement. Although real alerts from the backend were not available, the internal alert handling logic functioned correctly under test conditions.

4.3.3 Database Structure Verification

The SQLite database implemented through Drift was verified to correct creation and schema structure, but no fall events were populated through the RADIAN pipeline since the app is designed to receive entries from the Gotify server during integration. Instead, database functionality was validated by confirming that the database file was created successfully and that the Fall Log tab could render its UI elements, ensuring readiness for future integration.

4.3.4 Stability and Performance Testing

The application's performance was evaluated under repeated simulated alerts, theme changes, and navigation interactions. The app remained stable throughout testing, with no crashes or unexpected behavior, demonstrating that the subsystem is robust and ready for integration with the radar and data subsystems once backend communication is finalized.

4.4 Subsystem Conclusion

The Mobile Application Subsystem provides a functional, stable, and modular interface for caregivers, featuring alert simulation, user-configurable settings, theme controls, and a fully implemented local database designed for future fall event logging. While full integration with the Gotify server and backend pipeline has not yet been completed, the application developed correct UI behavior, stable performance, and readiness to receive and store real alerts once system integration is finalized.

5. Notification Server Subsystem Report

5.1 Subsystem Introduction

The Notification Server Subsystem plays a critical role in the mission success of the RADIAN system by providing a link between the Artificial Intelligence subsystem and the Mobile Application subsystem. Utilizing the Gotify server architecture, the Raspberry Pi will be able to communicate with the server by posting fall notifications to the server in real-time, allowing for quick and very accurate detection if a fall has occurred. Once the alert has been posted, the Mobile Application will then receive the notification, allowing the caretaker to act quickly to assist the fallen individual.

5.2 Subsystem Details

5.2.1 System Architecture

The Notification Server Subsystem serves as an intermediary between both the Data pipeline and the communication between the mmWave radar and the mobile application. Once the ML model has classified each point given by the mmWave radar and then detected a fall using the fall detection logic, an alert will be sent through a Python script on the Raspberry Pi. This Python script directly communicates with the Gotify server through the local IP address of the laptop that the RADIAN team tested on. An app token was also created to allow coordination between the Python script, the Gotify server, and the RADIAN mobile application. This architecture was chosen compared to a direct device-to-device connection to allow for increased reliability, scalability, and the possibility of incorporating multiple devices through the system.

5.2.2 Server Platform

The Notification Server Subsystem is built around Gotify, a lightweight, open-sourced notification server specifically designed for self-hosted environments. Gotify was selected due to its simple user interface and implementation, vast support throughout the Raspberry Pi community, and its compatibility with local networks, specifically how it interacts with the local IP address of devices. The backend of Gotify provides high performance while maintaining low memory and CPU overhead, allowing the machine learning to run simultaneously and not interfere with the server operations.

5.2.3 Raspberry Pi Integration

The Notification Server directly integrates with the Raspberry Pi, therefore serving as both the radar processing platform and the origin point for fall alerts. A simple, lightweight Python script publishes messages to the Gotify server using the standard HTTP POST requests, the standard for server communication in many servers across the globe. The token-based authentication needed for communication between the Pi and Gotify server eliminates the need for credential exchanges, allowing for the system to be both secure and simple to deploy.

5.3 Subsystem Validation

5.3.1 Server Deployment and Connectivity Testing

The deployment and connectivity between the Raspberry Pi and the local computer were tested through a “ping” between the two devices, verifying that both systems were able to communicate with each other through the local IP address of the testing laptop. The Gotify server was also downloaded and validated through inspection when viewing the “Pip List” of downloaded libraries on the Raspberry Pi device itself. The Gotify User Interface was also tested and validated through inspection on the laptop by running the Gotify software and opening the UI on a web browser through the IP address, allowing the RADIANT team to create and share the application token needed for testing.

5.3.2 Message Publishing and Payload Validation

The message publishing to the Gotify server through HTTP POST was validated by testing the Python script made by the RADIANT team. A simple message of “Hello World” was sent from the Raspberry Pi to the Gotify server and displayed on the Gotify UI as a message from the Raspberry Pi. An additional validation test was conducted to ensure that the server was able to handle incorrect tokens, where an error message was printed showing that the message was not correctly received.

5.3.3 Stability and Performance Testing

The subsystem’s performance was evaluated under repeated communications between the Raspberry Pi and the Gotify server. No alerts were dropped when testing multiple messages in rapid succession, and when the server was restarted, the service was still stable and able to receive messages.

5.4. Subsystem Conclusion

Overall, the design and validation of the Notification Server Subsystem demonstrate that it reliably fulfills its role within the broader RADIANT architecture. The Raspberry Pi can quickly and accurately send notifications to the Gotify server, ensuring that the caregiver can receive alerts in real time and assist the fallen individual. The subsystem is now ready for integration with the Mobile Application subsystem to communicate directly with the mobile application, removing the need for an intermediary laptop device.

6. Artificial Intelligence Subsystem Report

6.1 Subsystem Introduction

The Artificial Intelligence Subsystem plays a central role within the Data Subsystem by determining whether a fall event has occurred using a lightweight yet robust machine-learning pipeline. A Random Forest classifier, implemented in Python using scikit-learn, is used to distinguish static versus dynamic motion using synthesized radar point-cloud data. The classifier evaluates on multiple realistic features, including additive white Gaussian noise, jitter, and small movement shifts across the x-, y-, and z-axes, ensuring reliability under realistic conditions. A Python algorithm is then utilized to detect if a fall has occurred over the last five frames of the received data, printing a message if either a fall has occurred or not, depending on the various cases that were tested. Because this subsystem directly determines whether a fall is occurring, rigorous testing and cautious tuning are essential to minimize false positives while maintaining high sensitivity to true fall events.

6.2 Subsystem Details

6.2.1 System Architecture

The Artificial Intelligence Subsystem sits at the core of the RADIANT Data pipeline, as it processes all of the incoming radar point-cloud frames and then determines if a fall has occurred. The system architecture is based on two primary components: the random forest ML movement classifier and the Python-based fall detection algorithm. The radar data is received, processed, and then classified as either static or dynamic based on a variety of factors. These classifications are then fed into the fall detection algorithm, which determines if a fall has occurred over the last five frames. This design was chosen to balance adaptability with reliability over many different types of cases. The ML classifier provides robustness against noise and irregular behavior, while the fall detection algorithm reduces the amount of false positives by enforcing physical criteria for fall motion.

6.2.2 Machine Learning Model Design

The Machine Learning model is based on scikit-learn's random forest algorithm, as it is lightweight for the Raspberry Pi 3B+, but robust enough to resist overfitting and can handle noisy data. Because of concerns with damaging the only mmWave Radar board, a custom Python simulation pipeline was implemented to generate synthetic radar frames with varying motion patterns, Gaussian noise, jitter, and random movement shifts. The random forest evaluates dozens of different decision trees, each trained on slightly different subsets of the synthetic dataset. The result yields a stable and noise-tolerant classification between static and dynamic motion, enabling reliable performance under imperfect data.

6.2.3 Fall Detection Algorithm

Once each of the different motion states is classified, the fall detection algorithm constantly analyzes the five most recent frames to identify characteristic fall patterns. The algorithm checks for specific parameters to be true simultaneously, that the sustained dynamic movement is followed by a sudden stability and height reduction, to determine whether a fall occurred. This approach avoids any misclassifications caused by single noisy frames or any unwanted transient motions that may occur in realistic radar data. This fall detection is implemented as a lightweight Python script that runs locally on the Raspberry Pi. It displays an output message of either “Fall Detected” or “No Fall Detected” to use in the following subsystem pipeline, specifically the Notification Server Subsystem.

6.2.4 Raspberry Pi Integration

The Artificial Intelligence Subsystem is specifically designed to run on the Raspberry Pi's limited hardware environment. The random forest is designed to be smaller to ensure that inference times will remain well below the real-time requirements of constantly analyzing the data. All of the computations, including the point-cloud processing, ML inference, and fall logic evaluation, are completed within milliseconds of each frame input. The subsystem interfaces directly with the Data pipeline, running continuously without interfering with any other processes of the Pi or the pipeline as a whole. Memory usage is kept minimal as the Data team is not using a neural network architecture, making the system even more reliable during long-term duration.

6.3 Subsystem Validation

6.3.1 Model Training and Functional Verification

The functionality of the model was tested and verified by ensuring that the random forest classifier was correctly able to distinguish between the static and dynamic motion classes. The training accuracy and confusion matrix were inspected to verify that the classifier is reliably handling the realistic datasets of noise and jitter additions. The ML model was able to report an accuracy of 89% during the demo, passing the predetermined verification plan for the random forest model. An additional verification was conducted that ran the classifier on known synthetic test frames to confirm that the output labels matched the expected labels of the test frames. As the Data team continues to develop the model, they plan to continue to increase the accuracy to a possible 95%.

6.3.2 Edgecase Generation

To validate the fall logic algorithm, the Data team generated various edgecases that may cause false positives when running the algorithm. These edge cases include sitting, bending over, kneeling, lying down slowly, trip recovery, crawling, a pet walking across the frame, an object dropping, and a hard fall. To generate these edgecases, the Data team first defined what parameters would change from normal walking dynamic frames to the edgecases and then added noise and other jitter to demonstrate realistic data. Once these cases were generated, they were trained along with the normal radar data.

6.3.3 Frame Classification and Fall Logic Validation

To validate the end-to-end operation of the subsystem, the team executed the full radar-processing pipeline, from frame preprocessing, classification, and fall detection, using the various test cases that were implemented. These test cases include normal falls, static-only scenarios, and the various edgecases generated that were mentioned in **6.3.2**. For each test case, the AI subsystem printed the per-frame classification and the final fall vs. no fall decision. These outputs were then checked manually against the expected outcomes to verify the correctness of the ML model. An additional validation included intentionally corrupted frames and forcing impossible conditions to confirm that the fall detection logic failed safely without producing any false positives.

6.3.4 Stability and Performance Testing

The subsystem's stability was evaluated by running the classifier and fall detection algorithm continuously across hundreds of frames to ensure that the inference speed met real-time requirements on the Raspberry Pi. No performance degradation, memory delays, or classification delays were detected during its extended operation during demonstration.

6.4. Subsystem Conclusion

The Artificial Intelligence Subsystem successfully demonstrates the RADIAN system's ability to interpret radar data and determine whether or not a fall event has occurred with reliability and efficiency. Through the integration of the lightweight random forest classifier and a deterministic fall detection algorithm, the subsystem provides robust performance even under noisy conditions. These validation tests were essential in minimizing the false positives while maintaining sensitivity to true fall events. With its performance and modular design, the Artificial Intelligence Subsystem is now fully prepared for integration with the Notification Subsystem and Mobile Application Subsystem, enabling end-to-end fall detection for the overall RADIAN system.